NCERT Summary: Topic 09: Income & Employment Determination (Keynesian Model)

Source Integration

- NCERT Class 12 Introductory Macroeconomics, Chapter 4: Determination of Income and Employment

Core Factual & Conceptual Architecture

1. Conceptual Framework & Ex Ante vs. Ex Post

- Fixed Price and Interest Assumptions: The basic short-run Keynesian macroeconomic model assumes that general price levels and prevailing interest rates remain constant, meaning supply is infinitely elastic up to the capacity boundary.
- Ex Ante Values: Planned, desired, or intended economic magnitudes (such as planned consumption, planned investment, and expected sales) formulated by economic agents before actual market outcomes occur.
- Ex Post Values: Realized, actual, or accounting values recorded at the conclusion of an economic accounting period.
- Equilibrium Foundation: Equilibrium income determination is strictly governed by ex ante planned aggregates rather than ex post accounting identities.

2. Components of Aggregate Demand

- The Consumption Function:
  - Linear Consumption Equation: C = C_bar + cY
  - Autonomous Consumption (C_bar): Positive baseline consumption expenditure sustained even at zero income, funded via dissaving or asset liquidations.
  - Induced Consumption (cY): Income-dependent consumption expenditure driven by marginal earnings.
  - Marginal Propensity to Consume (MPC or c): The ratio of change in consumption to change in total income: Change in C / Change in Y. It strictly ranges between 0 and 1.
  - Marginal Propensity to Save (MPS or s): The ratio of change in saving to change in income: Change in S / Change in Y. Because total income is either consumed or saved, MPC + MPS = 1, meaning MPS = 1 - c.
  - Average Propensities: Average Propensity to Consume (APC = C / Y) and Average Propensity to Save (APS = S / Y), where APC + APS = 1.
- Investment Function:
  - Autonomous Investment (I_bar): In this basic baseline framework, investment outlays are treated as exogenous and fixed, independent of real income levels.

3. Equilibrium Determination & The Effective Demand Principle

- Aggregate Demand Formulation: AD = C + I = (C_bar + I_bar) + cY = A_bar + cY, where A_bar represents total autonomous expenditure.
- Effective Demand Equilibrium: Occurs where planned aggregate output (Y) strictly equals planned aggregate expenditure (AD):
  - Y = A_bar + cY, solving to Equilibrium Income: Y* = A_bar / (1 - c) = A_bar / s
- The Adjustment Mechanism (Unplanned Inventory Fluctuations):
  - When Planned Output exceeds Planned Demand (Y > AD): Goods remain unsold, triggering unintended, unplanned inventory accumulation. Firms react to inventory pileups by cutting output and labor in subsequent production cycles.
  - When Planned Demand exceeds Planned Output (AD > Y): Inventories are run down unexpectedly below desired buffer levels. Firms respond by hiring labor and expanding output.
  - Equilibrium Restored: Output adjusts until unintended inventory change strictly equals zero.

4. The Investment Multiplier

- Multiplier Concept: Demonstrates that an initial exogenous shift in autonomous spending (such as investment or government expenditure) generates a more than proportionate cumulative expansion in equilibrium national income.
- Mathematical Multiplier Derivation:
  - Multiplier (k) = Change in Income / Change in Autonomous Expenditure = 1 / (1 - c) = 1 / MPS
- Multiplier Dynamics: The higher the Marginal Propensity to Consume (c), the larger the multiplier effect, as a higher share of incremental income is continuously reinjected into successive rounds of consumption.

5. The Paradox of Thrift

- Definition: If all individuals across an economy attempt to raise their savings rate by increasing their Marginal Propensity to Save (MPS), total aggregate savings in the economy will either stagnate or decline.
- Operational Cause: An increase in MPS lowers the MPC, depressing aggregate demand. Reduced demand leads to unsold inventories, prompting firms to cut output. The resulting fall in national income reduces the absolute savings pool, neutralizing or worsening the initial effort to save more.

6. Output Gaps: Deficient vs. Excess Demand

- Full Employment Income (Y_f): The maximum output volume sustainable when all available economic resources are fully and efficiently deployed.
- Deficient Demand (Deflationary Gap):
  - Planned aggregate demand falls below the level required to maintain full employment output ($AD < Y_f$).
  - Economic Outcome: Involuntary unemployment, excess industrial capacity, and downward deflationary pressure.
- Excess Demand (Inflationary Gap):
  - Planned aggregate demand exceeds total output capacity at full employment ($AD > Y_f$).
  - Economic Outcome: Inability to increase physical production past full capacity, resulting in competitive price bidding and persistent demand-pull inflation.

✍ Notes & Updates